Problem 1. (Thursday, 4 points: 2+2)

a) Show that, if a simple closed (positively oriented) curve C is the boundary of a region R, then $\text{Area}(R) = \oint_C x\,dy = \oint_C -y\,dx$.

b) Use this result to calculate the area of the region between the x-axis and one arch of the cycloid with parametric equations $x = a(t - \sin t)$, $y = a(1 - \cos t)$. (Hint: the second line integral leads to easier calculations).

Problem 1, partie a:

Green's Theorem: $\oint_C M dx + N dy = \iint_R \left(\frac{\partial MM}{\partial xx} - \frac{\partial NN}{\partial yy}\right) dA$

Aire$(R) = \iint_R dA$ or Green's theorem $\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dA$

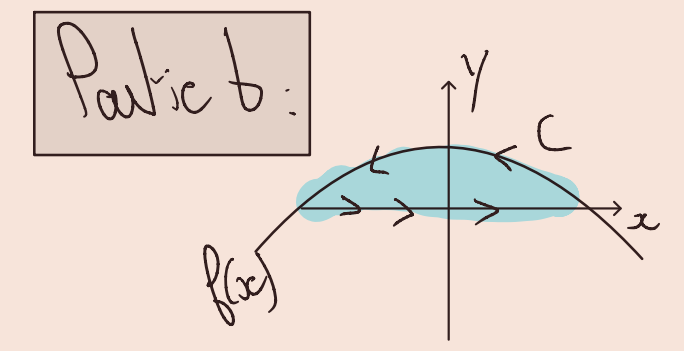
On a donc besoin que $\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) = 1$ pour avoir Area$(R) = \iint_R dA$

Pour $\oint_C x,dy$ ça donne $M=0$ et $N=x$, $\frac{\partial M}{\partial y} = 0$ et $\frac{\partial N}{\partial x} = 1 \iff \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 1-0$, on a bien Area$(R) = \iint_R (1-0) dA$

Pour $\oint_C y,dx$ ça donne $M=-y$ et $N=0$, $\frac{\partial M}{\partial y} = -1$ et $\frac{\partial N}{\partial x} = 0 \iff \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 0-(-1) = 1$, on a bien Area$(R) = \iint_R (0-(-1)) dA$

Et Area$(R) = \oint_C x,dy = \oint_C y,dx$

Partie b:



Deux parties, la première le long de l'axe x et la seconde le long du cycloïde

En utilisant $\oint_C -y,dx$ vaut 0 sur la première partie, donc l'aire se résume à : Area $= \oint_C -y,dx = \oint_{cycloid} -y,dx$

$f(x) = y = a(1-\cos t)$
$x = a(t - \sin t)$

$x = a(t - \sin t)$ $dx = a - a\cos t\, dt$

$\int_{cycloide} -y,dx = \int -a(1-\cos t)\cdot a(1-\cos t) dt$

Cycloïde $=0$ quand $y=0$
donc $a(1-\cos t) = 0$
$\begin{cases} t=0, \cos t\ 1 \\ \text{ou} \\ t = 2\pi, \cos t = 1 \end{cases}$

$\int_0^{2\pi} -a(1-\cos t)\cdot a(1-\cos t) dt$

$\int_0^{2\pi} -a^2(1-\cos t)^2 dt$

$-a^2 \int (1-\cos t)^2 dt$ $(A-B)^2 = a^2 - 2ab + b^2$

$-a^2 \int_0^{2\pi 0} 1^2 - 2\cos t + (\cos t)^2 dt$ $\cos^2 t = \frac{1+\cos 2t}{2}$

$\cos(2t)dt = -2\sin(2t)$

$-a^2 \int_0^{2\pi} t - 2\sin t + \frac{1}{2}\left(t + \frac{\sin(2t)}{2}\right)$

$-a^2 \left[t - 2\sin(t) + \frac{1}{2}\left(t + \frac{\sin(2t)}{2}\right)\right]_0^{2\pi}$

$-a^2 \left[2\pi - 2\underbrace{\sin(2\pi)}_{=0} + \frac{1}{2}\left(2\pi + \frac{\overbrace{\sin(4\pi)}^{=0}}{2}\right)\right] - 0$

$-a^2 [2\pi + \pi] = -3a^2\pi$

Problem 3. (Friday, 6 points: 2+2+2)

a) Let C be the unit circle, oriented counterclockwise, and consider the vector field $\vec{F} = xy\,\hat{\imath} + y^2\,\hat{\jmath}$. Which portions of C contribute positively to the flux of $\vec{F}$? Which portions contribute negatively?

b) Find the flux of $\vec{F}$ through C by direct calculation (evaluating a line integral). Explain your answer using (a).

c) Find the flux of $\vec{F}$ through C using Green's theorem.

Partie a: Flux $= \oint \vec{F}\cdot\hat{n} ds$

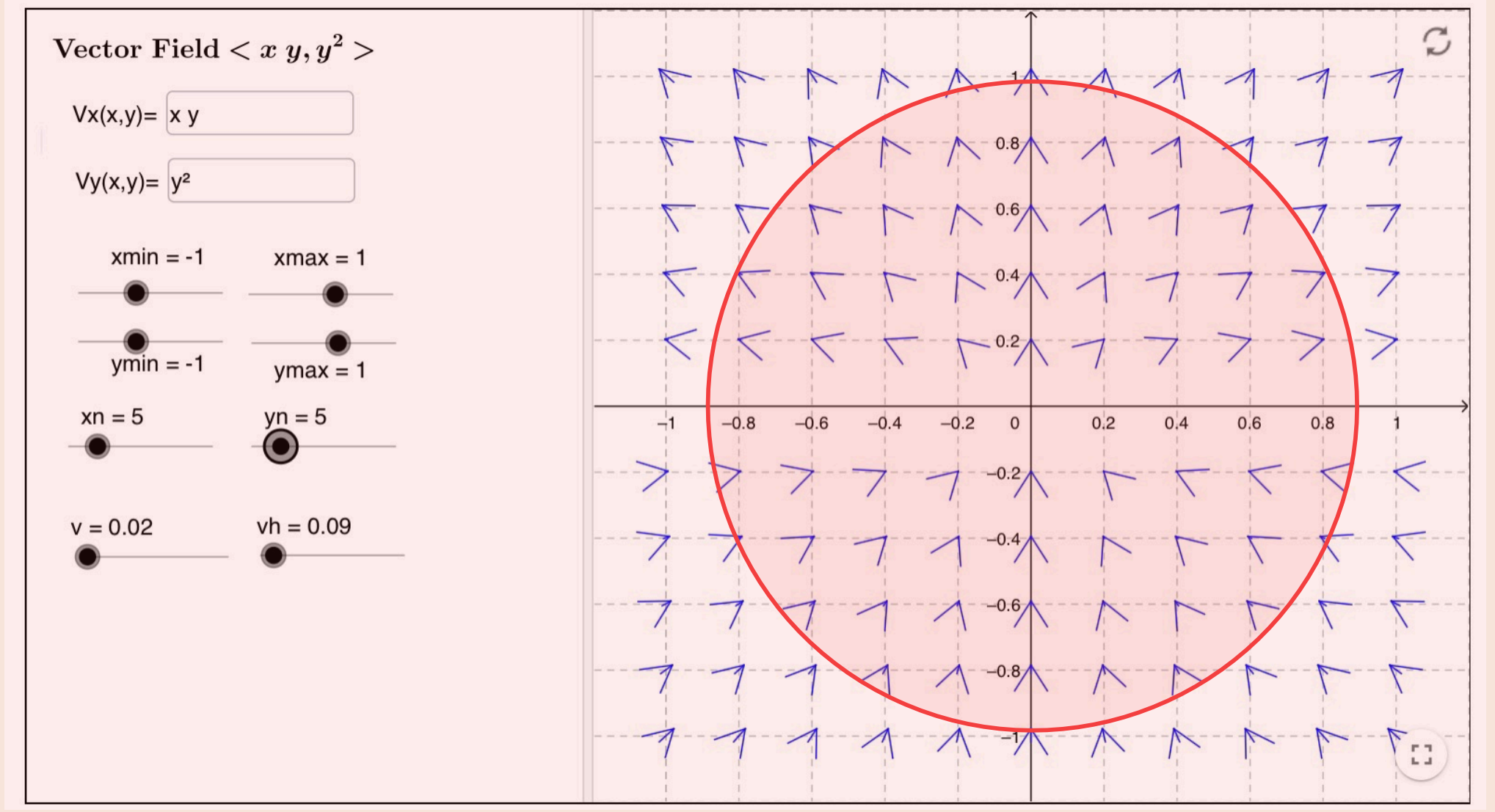
Flux positif quand $\vec{F}\cdot\hat{n} > 0$ donc $(xy, y^2)\cdot(x, y) = (x^2y + y^3) = y\underbrace{(x^2+y^2)}$

Sur le cercle unité c'est égal à 1.

Donc le signe du flux dépend du signe de y

Flux positif si $y > 0$ (demi plan supérieur)

Flux négatif si $y < 0$ (demi plan inférieur)



Vector Field $< x\,y, y^2 >$

Vx(x,y)= x y

Vy(x,y)= y²

xmin = -1

xmax = 1

ymin = -1

ymax = 1

xn = 5

yn = 5

v = 0.02

vh = 0.09

Partie b: $x = \cos t$ $y = \sin t$ $t \in [0, 2\pi]$

Flux $= \oint_C -Qdx + Pdy$ avec $\vec{F} = P\hat{\imath} + Q\hat{\jmath}$ $P = xy$ et $Q = y^2$

$dx = -\sin t$ $dy = \cos t$

$P = \cos t \sin t$

$Q = \sin^2 t$

$\oint_0^{2\pi} -\sin^2 t(-\sin t) + \cos t \sin t (\cos t) dt = \int_0^{2\pi} \sin^3 t + \cos^2 t \sin t\, dt$

$= \int_0^{2\pi} \sin t \underbrace{\left(\sin^2 t + \cos^2 t\right)}_{=1} dt = \int_0^{2\pi} \sin t\, dt = \int_0^{2\pi} -\cos t = [-\cos t]_0^{2\pi} = -1-(-1) = 0$ Logique la moitié supérieure contribue positivement et la moitié inférieure contribue négativement

Partie c: Idem mais avec Green's Theorem

$\oint_C \vec{F}\cdot\hat{n} ds = \iint \underbrace{\text{div}(\vec{F})}_{\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}} dA = \iint y + 2y\, dA = \iint 3y\, dA = \iint 3\sin\theta\, r, dr, d\theta = \iint 3r\sin\theta\, dr, d\theta = \int_0^{2\pi}\int_0^1 \frac{3r^2}{2}\sin\theta\, d\theta = \int_0^{2\pi} \frac{3\sin\theta}{2} d\theta = \int_0^{2\pi} -\frac{3}{2}\cos\theta = \left[-\frac{3}{2}\cos\theta\right]_0^{2\pi} = \left[-\frac{3}{2}\right] - \left[-\frac{3}{2}\right] = -\frac{3}{2} + \frac{3}{2} = 0$

Problem 2. (Thursday, 6 points: 4+2)

a) For what simple closed (positively oriented) curve C in the plane does the line integral $\oint_C (x^2y + y^3 - y)\,dx + (3x + 2y^2x + e^y)\,dy$ have the largest positive value? (use Green's theorem).

b) Determine what this value is.

Partie a:

$\oint_C \underbrace{(x^2y + y^3 - y)}_{M} dx + \overbrace{(3x + 2y^2x + e^y)}^{N} dy = \iint_R (N_x - M_y) dA$

$M_y = x^2 + 3y^2 - 1$ $N_x = 3 + 2y^2$ $= \iint_R \left(3 + 2y^2 - (x^2 + 3y^2 - 1)\right) dA$

$= \iint_R (4 - y^2 - x^2) dA$

On cherche à maximiser $4 - y^2 - x^2 \iff 4 - \overbrace{(y^2 + x^2)}^{\text{cercle}}$ donc $(y^2 + x^2) \leq 4 \iff y^2 + x^2 \leq 2$

Donc C est le cercle centré à l'origine de rayon 2.

Partie b: Pour intégrer autant passer par les coordonnées polaires

$x = r\cos\theta$ $x^2 + y^2 = r^2$ $dA = r, dr, d\theta$

$y = r\sin\theta$

$\iint_R 4 - (y^2 + x^2) dA \iff \iint_R 4 - (r^2) r, dr, d\theta \iff \iint_R 4r - r^3 dr, d\theta$

$\int_0^{2\pi}\int_0^2 4r - r^3 dr d\theta \iff \int_0^{2\pi}\int_0^2 \frac{4r^2}{2} - \frac{r^4}{4} d\theta \iff \int_0^{2\pi} \left[\frac{4r^2}{2} - \frac{r^4}{4}\right]_0^2 d\theta \iff \int_0^{2\pi} \left[\frac{16}{2} - \frac{16}{4}\right] d\theta \iff \int_0^{2\pi} 4, d\theta \iff \int_0^{2\pi} 4\theta \iff 8\pi$